MILITARY SPECIFICATION SHEET

NUT, CAPTIVE WASHER, SELF-LOCKING, NICKEL ALLOY 718, 220 KSI Ftu, 900° (12-Spline) WRENCHING ELEMENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the nut described herein shall consist of this document and the latest issue of MIL-N-85730.

1. DESIGN SELECTION: These nuts are designed for use with expandable sleeve bolts conforming to MIL-B-85667.

2. APPLICATION CRITERIA: The nuts shall be used in accordance with the limitations of MS33588.

3. TOOLING: Per MIL-W-8982

4. GENERAL REQUIREMENTS:

4.1 Material:

4.1.1 Nut: Nickel alloy (Inconel 718) per AMS 5662.

4.1.2 Washer: Cres (A-286) per AMS 5525

4.2 Heat Treatment:

4.2.1 Nut: Heat treat to 220 KSI min per MIL-H-6875.

4.2.2 Washer: Heat treat to 140 KSI min per MIL-H-6875.

4.3 Surface Treatment:

4.3.1 Nut: Passivate per QQ-P-35.

4.3.2 Washer: Passivate per QQ-P-35.

4.4 Lubricant:

4.4.1 Nut: Dry Film lubricate per MIL-L-8937

Cetyl alcohol lubricate per MIL-L-87132.

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4.4.2 Washer: Dry film lubricate per MIL-L-8937.

Cetyl alcohol lubricate per MIL-L-87132.

4.5 Workmanship:

4.5.1 Dimensions are in inches.

4.5.2 Dimensions apply before lubrication.

4.5.3 Tolerances: As specified in Figure 1 and Table I.

4.5.4 Bearing Surface: Bearing surface shall be normal with pitch diameter of thread within the Y values specified, see Table I.

4.5.5 Concentricity: N diameter shall be concentric to pitch diameter of thread with X values T.I.R., see Table I.

4.6 Example of military part number:

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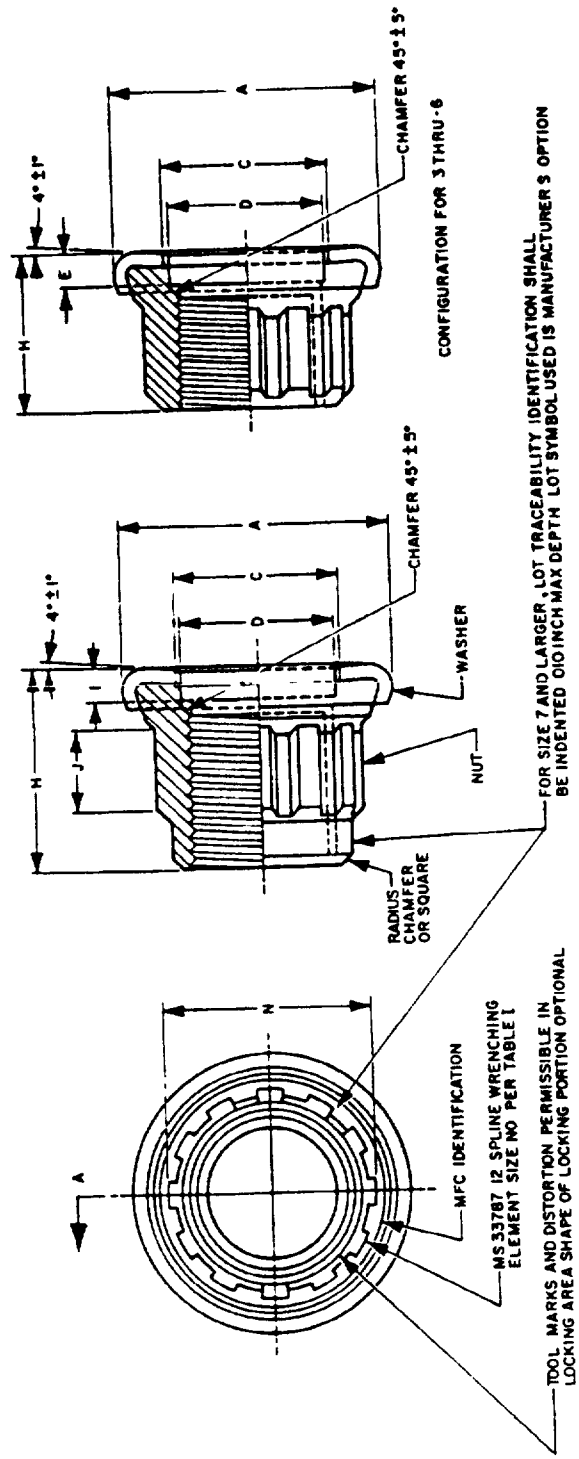
- 4 — Thread: 0.250 - 28 UNJF-3B
- - — Lubricant: "-" denotes non-dry film lubricant. "L" denotes dry film lubricant in accordance with MIL-L-8937.
- M85730/1 — Basic Specification Sheet Number

5. ADMINISTRATIVE AND CONTRACTURAL PROVISIONS:

5.1 For design feature purposes this document takes precedence over procurement documents referenced herein.

5.2 Referenced documents shall be of the issue in effect on date of invitation for bid.

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SECTION-A

FIGURE 1

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TABLE I Nut Dimensions.

Size Dash No.	Thread Size MIL-S-8879	MS33787 (12 Spline) Element Number	A Max	C ±.005	D ±.005	E +.010 - 005
3	.1900-32UNJF-3B	8	.460	.288	.282	.077
4	.2500-28UNJF-3B	10	.560	.353	.347	.077
5	.3125-24UNJF-3B	12	.660	.418	.412	.077
6	.3750-24UNJF-3B	14	.760	.485	.479	.077
7	.4375-20UNJF-3B	18	.860	.552	.546	.077
8	.5000-20UNJF-3B	20	1.000	.636	.621	.077
9	.5625-18UNJF-3B	22	1.100	.765	.750	.087
10	.6250-18UNJF-3B	24	1.310	.840	.825	.087
12	.7500-16UNJF-3B	30	1.480	.987	.972	.092
14	.8750-14UNJF-3B	34	1.690	1.111	1.096	.097
16	1.0000-12UNJF-3B	38	1.834	1.253	1.238	.097
18	1.1250-12UNJF-3B	44	2.100	1.425	1.410	097
20	1.2500-12UNJF-3B	48	2.250	1.500	1.485	.097
22	1.3750-12UNJF-3B	52	2.480	1 650	1 635	.097
24	1.5000-12UNJF-3B	56	2 690	1.780	1.765	.097

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Size	H		J	N	X	Y	Ultimate Axial Tensile Strength Lb Min (a)
	Max	Min	Min	Max			
3	.280	.260	.080	.308	.003	.004	4,970
4	.332	.312	.100	.381	.003	.004	8,890
5	.400	.380	.120	.456	.003	.005	14,100
6	.478	.458	.152	.531	.003	.006	20,900
7	.525	.505	.168	.679	.003	.008	28,300
8	.603	.583	.190	.753	.003	.008	37,800
9	.672	.652	.230	.828	.004	.009	47,900
10	734	.714	.250	.901	.004	.010	59,900
12	.879	.859	.310	1.124	.004	.012	86,900
14	1.012	.992	.350	1.270	.005	.014	119,000
16	1.140	1.120	.400	1.419	.006	.016	155,000
18	1.337	1.317	450	1.641	.007	.019	198,000
20	1.524	1.504	.500	1.788	.007	.021	247,000
22	1.659	1.639	.550	1.937	.008	.023	301,000
24	1.847	1.827	.600	2.085	.008	.023	361,000

(a) Axial tensile strength determined from formula Wa = FtuA. Where. A is the cross sectional area in square inches, based on the maximum pitch diameter of bolt thread, Ftu is 220 KSI and Wa is the axial tensile strength in pounds.

Custodians:
Army - AV
Navy - AS
Air Force -11

Review activities:
DLA-IS

Preparing activity:
Navy - AS
(Project No. 5310-1426)

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